

DeepMiner™ DV-3100 Subsurface Haul Truck

OEM Field Manual — Volume 4

AI-Augmented Field Engineer Edition

SECTION 1 — SAFETY & COMPLIANCE

1.1 General Safety

- Only certified underground-operations personnel may operate or service the DV-3100.
- Perform Lockout/Tagout before accessing suspension, drivetrain, or load-bed components.
- Maintain a minimum 5 m clearance behind the haul truck during reversing operations.
- Wear Class-V mining PPE: respirator, reinforced boots, high-visibility underground gear.

1.2 Hazard Zones

- **Red Zone:** Rear haul bed (crush hazard during dumping).
- **Orange Zone:** Suspension cluster (high-compression hazard).
- **Yellow Zone:** Engine compartment (thermal and fume hazard).

1.3 AI-Assisted Safety

The AI Field Engineer Tool automatically:

- Predicts suspension imbalance using terrain-response analytics.
 - Detects load-weight anomalies that may cause rollover risk.
 - Flags drivetrain overheating conditions before critical thresholds.
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SECTION 2 — SYSTEM OVERVIEW

2.1 Machine Description

The DeepMiner™ DV-3100 is a heavy-duty subsurface haul truck engineered for transporting ore, debris, and structural materials through narrow underground corridors. Its terrain-adaptive suspension and high-traction drivetrain ensure stability on uneven mine floors.

2.2 Core Subsystems

- **Suspension System:** QS-3100 dampers + load-balancing struts

- **Powertrain:** TD-900 TurboDiesel engine + TR-MegaTorque transmission
 - **Load System:** Reinforced haul bed with RockShield coating
 - **Sensor Array:** Suspension imbalance, terrain mapping, load weight
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SECTION 3 — BILL OF MATERIALS (BOM)

3.1 Structural Assembly

- Reinforced Haul Bed (RockShield-Coated)
- Heavy-Duty Frame (DV-Series)
- Operator Cab (Pressurized)

3.2 Suspension System

- QuadShock QS-3100 Dampers
- Load-Balancing Struts
- Terrain Response Linkages
- Suspension Control Module (SCM-3)

3.3 Powertrain

- TurboDiesel Engine TD-900
- Transmission TR-MegaTorque
- Differential Set (High-Traction)
- Cooling System (Mine-Grade)

3.4 Sensor & Control

- Suspension Imbalance Sensors (SIS-4)
- Terrain Mapping Array (TMA-2)
- Load Weight Sensors (LWS-5)
- Cab Console (Dust-Sealed)

3.5 Consumables

- Shock Fluid QS-Grade
 - Engine Oil TD-Max
 - Brake Pads (Heat-Resistant)
 - Air Scrubber Filters
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SECTION 4 — OPERATING PROCEDURES

4.1 Startup Sequence

1. Inspect haul bed for structural cracks.

2. Power on TD-900 engine.
3. Run **AI-Assisted Suspension Calibration** (12 seconds).
4. Engage drivetrain.
5. Begin controlled haul operations.

4.2 Hauling Workflow

- Maintain load weight within 85% of rated capacity.
 - AI flags terrain instability before entering narrow corridors.
 - Use low-gear mode for steep inclines or wet mine floors.
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SECTION 5 — FIELD DIAGNOSTICS

5.1 Common Failures

- Suspension imbalance
- Drivetrain overheating
- Load distribution errors
- Terrain-response lag

5.2 AI-Assisted Diagnostics

The AI tool provides:

- Suspension imbalance heatmaps
- Drivetrain thermal stress graphs
- Load distribution modeling
- Terrain-response prediction curves

5.3 Field Repair Procedure: Correcting Suspension Imbalance

1. Activate **AI Suspension Scan**.
 2. AI identifies imbalance source (QS-3100 damper, strut, linkage, terrain mismatch).
 3. Inspect QS-3100 damper for compression deviation >18%.
 4. Replace load-balancing strut if structural wear exceeds 30%.
 5. Realign terrain-response linkages.
 6. Re-run suspension scan to confirm stability.
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SECTION 6 — MAINTENANCE

6.1 Daily Maintenance

- Inspect suspension components.

- Check haul bed hinge points.
- Clean sensor array.

6.2 Weekly Maintenance

- Replace air scrubber filters.
- Inspect drivetrain cooling system.
- Run AI-guided **Suspension Health Scan**.

6.3 Monthly Maintenance

- Replace QS-3100 shock fluid.
- Perform full terrain-response calibration.
- Update firmware via AI Fleet Console.

SECTION 7 — TROUBLESHOOTING MATRIX

Issue	Likely Cause	AI Diagnostic Output	Field Fix
Suspension Imbalance	Damper wear	Imbalance heatmap	Replace QS-3100 damper
Drivetrain Overheat	Cooling failure	Thermal stress graph	Inspect cooling system
Load Errors	Uneven distribution	Load modeling alert	Rebalance haul bed
Terrain Lag	Sensor drift	Response curve deviation	Recalibrate TMA-2 array

SECTION 8 — TECHNICAL SPECIFICATIONS

- **Engine Power:** 900 HP
- **Load Capacity:** 32 tons
- **Suspension Compression:** 3100-Series adaptive
- **Operating Temp:** -20°C to 55°C
- **Weight:** 18,400 kg
- **Max Gradeability:** 28%